PAAVANI KARUTURI

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SUMMARY

Software Engineer with 3+ years building production backend, distributed, and AI/ML systems using Python, Java, Go, TypeScript, cloud infrastructure, and modern ML frameworks. Experienced shipping customer-facing features, LLM/RAG workflows, scalable APIs, automated testing, observability, and performance optimizations from design through production.

TECHNICAL SKILLS

Programming: Python, Java, Go, C++, TypeScript, JavaScript, SQL, Bash, Linux
AI & Machine Learning: PyTorch, TensorFlow, Hugging Face, LLMs, RAG, embeddings, LoRA, model evaluation, vector search
Backend & Distributed Systems: FastAPI, Spring Boot, Node.js, REST APIs, GraphQL, Kafka, Redis, microservices, concurrency
Cloud & Data: AWS, GCP, PostgreSQL, MySQL, MongoDB, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD
Observability & Quality: Prometheus, Grafana, OpenTelemetry, Jest, Cypress, regression testing, performance profiling

PROFESSIONAL EXPERIENCE

TipTop 360, San Jose, CA | **Full Stack Software Engineer** May 2026 – Present

- Built customer-facing workflows with **React, TypeScript, Node.js, GraphQL, and REST** for booking, refund, and policy operations.
- Shipped **LangGraph/RAG agents** with streaming outputs and idempotent tools, reducing manual information lookups by **30%**.
- Fine-tuned an open-source transformer with **LoRA, PyTorch, and Hugging Face** after prompt, context, and tool-schema optimization.
- Benchmarked Claude, Gemini, Llama, and Mistral on latency, grounding, and failure modes to guide production model selection.
- Tuned batching, caching, async execution, and retrieval paths, cutting latency from **8s to 5s** and user-reported issues **40%**.

Olympus, San Jose, CA | **Backend Engineer Intern** May 2025 – Aug 2025

- Engineered a production **Python** microservice with secure REST APIs, RBAC, and request validation supporting **200+ users**.
- Instrumented **Prometheus** health and p95 telemetry, isolating hot paths and reducing API latency **25%** with zero regressions.
- Reproduced failures across auth, service, and data layers; added edge-case tests to prevent regressions and improve reliability.
- Automated provisioning handoffs with audit logs and structured errors, streamlining account provisioning by **25%**.

Fractal Analytics, Mumbai, India | **Cloud and Data Tech Engineer** Jun 2023 – Jul 2024

- Delivered **12 production features** in Python, TypeScript, and Java across backend and customer-facing platform workflows.
- Eliminated **60%** redundant computation through **PostgreSQL indexing** and **Redis caching** on high-volume read paths.
- Reduced hot-path query latency roughly **33%** by profiling execution plans and tuning database access patterns.
- Expanded **Jest/Cypress** coverage **40%** with **GitHub Actions** regression gates across UI and API workflows.
- Reduced production regressions **30%** through automated UI/API validation and repeatable CI/CD release checks.

Rivach, Hyderabad, India | **Backend Engineer** Mar 2022 – Jun 2023

- Engineered **Java REST APIs** serving **25K daily users** and sustaining high-frequency write workflows across backend services.
- Designed **MySQL** schemas and transaction paths connecting persistent storage with Java services and stable API contracts.
- Diagnosed race conditions and concurrency failures under high-frequency writes across service and database boundaries.
- Tuned query plans and data-access paths, reducing p95 latency roughly **30%** under concurrent load.
- Automated Docker deployments with **GitHub Actions**, increasing deployment frequency **40%** with zero-downtime releases.

PROJECTS

Enterprise Document Intelligence Platform | **GitHub**

- Engineered a multimodal document pipeline with **Python, FastAPI, PyTorch, and Hugging Face** across **1K+ document pages**.
- Evaluated extraction across scans, forms, tables, and PDFs, achieving **90%+ field-level F1** on labeled document test sets.
- Optimized preprocessing and inference paths to achieve **sub-800 ms p95 latency** while serving **30+ concurrent requests**.
- Instrumented **Prometheus/Grafana** and **100+ regression tests** for latency, confidence, failures, and model-quality monitoring.

AegisFlow – High-Throughput Distributed API Platform

- Engineered high-throughput **Go** services and optimized concurrent execution paths, sustaining approximately **58K requests/sec**.
- Containerized services with **Docker/Kubernetes** and integrated Redis for low-latency distributed request handling.
- Instrumented services with **OpenTelemetry** and **Prometheus** to expose latency, runtime behavior, and operational health.
- Profiled concurrency and request paths under load to identify bottlenecks and improve service efficiency and reliability.

CERTIFICATIONS & ACHIEVEMENTS

- AWS Certified Solutions Architect – Associate
- Google Cloud Computing Foundations
- Google Cloud: Generative AI with Vertex AI
- Runner-up – Stanford LLM x Law Hackathon (2025)
- Winner – Cloud Native Hackathon (2022)
- Microsoft Azure Cloud Services

EDUCATION

San Jose State University, San Jose, CA | **Masters in Software Engineering** Aug 2024 – May 2026
Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Engineering, NLP, Deep Learning, Data Mining, Cloud Computing